

Kubernetes Task

Task Description:

Setup minikube at your local and explore creating namespaces (Go through official documentation).

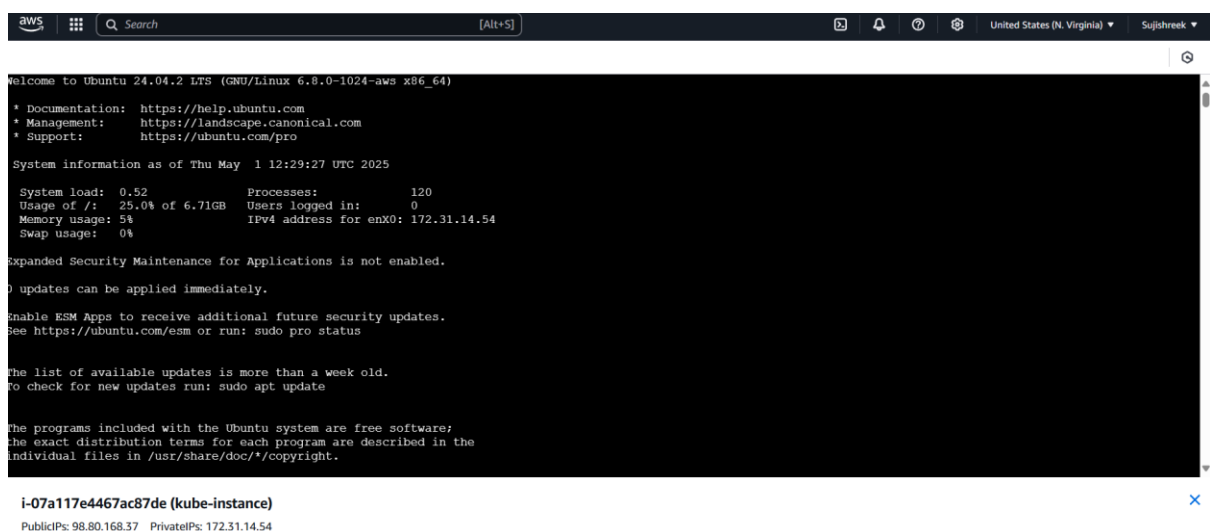
Techstacks needs to be used :

- Vbox, WSL
- Docker
- Minikube
- Kubectl

If the Local system has any issues you can use AWS.

- AWS EC2 (t2.medium)

SCREENSHORTS:



The screenshot shows an AWS console terminal window with the following content:

```
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1024-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Thu May 1 12:29:27 UTC 2025

System load:  0.52               Processes:    120
Usage of /:   25.0% of 6.71GB    Users logged in:  0
Memory usage: 5%                IPv4 address for enx0: 172.31.14.54
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

i-07a117e4467ac87de (kube-instance)
PublicIPs: 98.80.168.37 PrivateIPs: 172.31.14.54

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United States (N. Virginia)

Sujishreek

ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

ro run a command as administrator (user "root"), use "sudo <command>". See "man sudo_root" for details.

ubuntu@ip-172-31-14-54:~\$ sudo apt update && sudo apt upgrade -y

Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease

Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]

Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]

Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]

Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]

Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]

Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]

Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]

Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]

Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [1118 kB]

Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]

Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]

Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1028 kB]

Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [782 kB]

Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [224 kB]

Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]

Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [13.5 kB]

Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1059 kB]

Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [260 kB]

Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [376 kB]

Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [26.0 kB]

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Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [468 B]

Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.6 kB]

Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3792 B]

Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]

Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]

Fetch: 33.7 MB in 5s (6331 kB/s)

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

2 packages can be upgraded. Run 'apt list --upgradable' to see them.

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

Calculating upgrade... Done

The following NEW packages will be installed:

linux-aws-headers-6.8.0-1027 linux-aws-tools-6.8.0-1027 linux-headers-6.8.0-1027-aws linux-image-6.8.0-1027-aws linux-modules-6.8.0-1027-aws

linux-tools-6.8.0-1027-aws

The following upgrades have been deferred due to phasing:

fwupd grub-common grub-efi-amd64-bin grub-efi-amd64-signed grub-pc grub-pc-bin grub2-common libfwupd2

The following packages will be upgraded:

apparmor apport apport-core-dump-handler cloud-init dirmngr distro-info-data gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpgconf gpgsm gpgv keyboard-landscape-common libapparmor1 libarchive13t64 libbwt64 libelft64 libexpat1 liblzma5 libnetplan1 libnss-systemd libpam-systemd libperl5.38t64 libplymouth5 libsystemd-shared libudev1 libxml2 libxslt1.1 linux-aws linux-base linux-headers-aws linux-image-aws linux-tools-common netplan-generator netplan.io openssh-client openssh-server openssh-sftp-server pci.ids perl perl-base perl-modules-5.38 plymouth plymouth-themes-ubuntu-text python3-apport python3-jinja2 python3-netplan python3-problem-report python3-software-properties rsyslog snapd software-properties-common sosreport systemd systemd-dev systemd-resolved systemd-sysv tzdata tzdata-legacy ubuntu-pro-client ubuntu-pro-client-l10n udev update-notifier-common vim vim-common vim-runtime vim-tiny xxd xz-utils

4 upgraded, 6 newly installed, 0 to remove and 8 not upgraded.

8 standard LTS security updates

Need to get 143 MB of archives.

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Need to get 143 MB of archives.

After this operation, 185 MB of additional disk space will be used.

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libperl5.38t64 amd64 5.38.2-3.2ubuntu0.1 [4871 kB]

Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl amd64 5.38.2-3.2ubuntu0.1 [231 kB]

Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-base amd64 5.38.2-3.2ubuntu0.1 [1827 kB]

Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-modules-5.38 all 5.38.2-3.2ubuntu0.1 [3110 kB]

Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libnss-systemd amd64 255.4-1ubuntu8.6 [159 kB]

Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-dev all 255.4-1ubuntu8.6 [104 kB]

Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-resolved amd64 255.4-1ubuntu8.6 [296 kB]

Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd-shared amd64 255.4-1ubuntu8.6 [2073 kB]

Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd0 amd64 255.4-1ubuntu8.6 [433 kB]

Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-sysv amd64 255.4-1ubuntu8.6 [11.9 kB]

Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-systemd amd64 255.4-1ubuntu8.6 [235 kB]

Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd amd64 255.4-1ubuntu8.6 [3471 kB]

Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 udev amd64 255.4-1ubuntu8.6 [1873 kB]

Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libudev1 amd64 255.4-1ubuntu8.6 [175 kB]

Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapparmor1 amd64 4.0.1really4.0.1-0ubuntu0.24.04.4 [50.6 kB]

Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 liblzma5 amd64 5.6.1+really5.4.5-1ubuntu0.2 [127 kB]

Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-problem-report all 2.28.1-0ubuntu3.5 [24.9 kB]

Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-apport all 2.28.1-0ubuntu3.5 [92.4 kB]

Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apport-core-dump-handler all 2.28.1-0ubuntu3.5 [17.9 kB]

Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apport all 2.28.1-0ubuntu3.5 [84.7 kB]

Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-server amd64 1:9.6p1-3ubuntu13.11 [37.3 kB]

Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-server amd64 1:9.6p1-3ubuntu13.11 [509 kB]

Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-client amd64 1:9.6p1-3ubuntu13.11 [905 kB]

Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 update-notifier-common all 3.192.69.1 [207 kB]

Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 gpg-wks-client amd64 2.4.4-2ubuntu17.2 [70.9 kB]

Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 dirmngr amd64 2.4.4-2ubuntu17.2 [323 kB]

Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 gnupg-utils amd64 2.4.4-2ubuntu17.2 [109 kB]

i-07a117e4467ac87de (kube-instance)

PublicIPs: 98.80.168.37 PrivateIPs: 172.31.14.54

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Get:76 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 software-properties-common all 0.99.49.2 [14.4 kB]
Get:77 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-software-properties all 0.99.49.2 [29.8 kB]
Get:78 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 snapd amd64 2.67.1+24.04 [30.1 MB]
Get:79 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 sosreport amd64 4.8.2-0ubuntu0-24.04.1 [364 kB]
Get:80 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 cloud-init all 24.4.1-0ubuntu0-24.04.3 [604 kB]
Fetched 143 MB in 3s (54.4 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
(Reading database ... 70560 files and directories currently installed.)
Preparing to unpack .../libperl5.38t64_5.38.2-3.2ubuntu0.1_amd64.deb ...
Unpacking libperl5.38t64:amd64 (5.38.2-3.2ubuntu0.1) over (5.38.2-3.2build2.1) ...
Preparing to unpack .../perl_5.38.2-3.2ubuntu0.1_amd64.deb ...
Unpacking perl (5.38.2-3.2ubuntu0.1) over (5.38.2-3.2build2.1) ...
Preparing to unpack .../perl-base_5.38.2-3.2ubuntu0.1_amd64.deb ...
Unpacking perl-base (5.38.2-3.2ubuntu0.1) over (5.38.2-3.2build2.1) ...
Setting up perl-base (5.38.2-3.2ubuntu0.1) ...
(Reading database ... 70560 files and directories currently installed.)
Preparing to unpack .../0-perl-modules-5.38_5.38.2-3.2ubuntu0.1_all.deb ...
Unpacking perl-modules-5.38 (5.38.2-3.2ubuntu0.1) over (5.38.2-3.2build2.1) ...
Preparing to unpack .../1-libnss-systemd_255.4-1ubuntu8.6_amd64.deb ...
Unpacking libnss-systemd:amd64 (255.4-1ubuntu8.6) over (255.4-1ubuntu8.5) ...
Preparing to unpack .../2-systemd-dev_255.4-1ubuntu8.6_all.deb ...
Unpacking systemd-dev (255.4-1ubuntu8.6) over (255.4-1ubuntu8.5) ...
Preparing to unpack .../3-systemd-resolved_255.4-1ubuntu8.6_amd64.deb ...
Unpacking systemd-resolved (255.4-1ubuntu8.6) over (255.4-1ubuntu8.5) ...
Preparing to unpack .../4-libsystemd-shared_255.4-1ubuntu8.6_amd64.deb ...
Unpacking libsystemd-shared:amd64 (255.4-1ubuntu8.6) over (255.4-1ubuntu8.5) ...
Preparing to unpack .../5-libsystemd0_255.4-1ubuntu8.6_amd64.deb ...
Unpacking libsystemd0:amd64 (255.4-1ubuntu8.6) over (255.4-1ubuntu8.5) ...

i-07a117e4467ac87de (kube-instance)

PublicIPs: 98.80.168.37 PrivateIPs: 172.31.14.54

aws

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buntu@ip-172-31-14-54:~\$ sudo systemctl start docker
buntu@ip-172-31-14-54:~\$ sudo usermod -aG docker \$USER
buntu@ip-172-31-14-54:~\$ newgrp docker
buntu@ip-172-31-14-54:~\$ docker run hello-world
Unable to find image 'hello-world:latest' locally
atest: Pulling from library/hello-world
6590344b1a5: Pull complete
Digest: sha256:c41088499908a59aae84b0a49c70e86f4731e588a737f1637e73c8c09d995654
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the
executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
to your terminal.

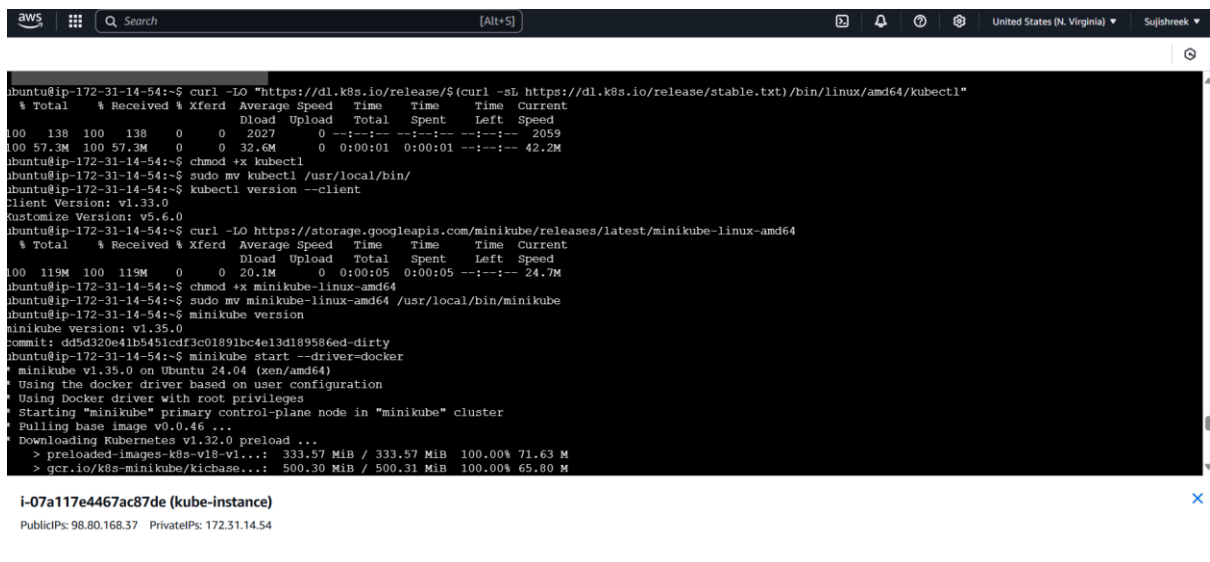
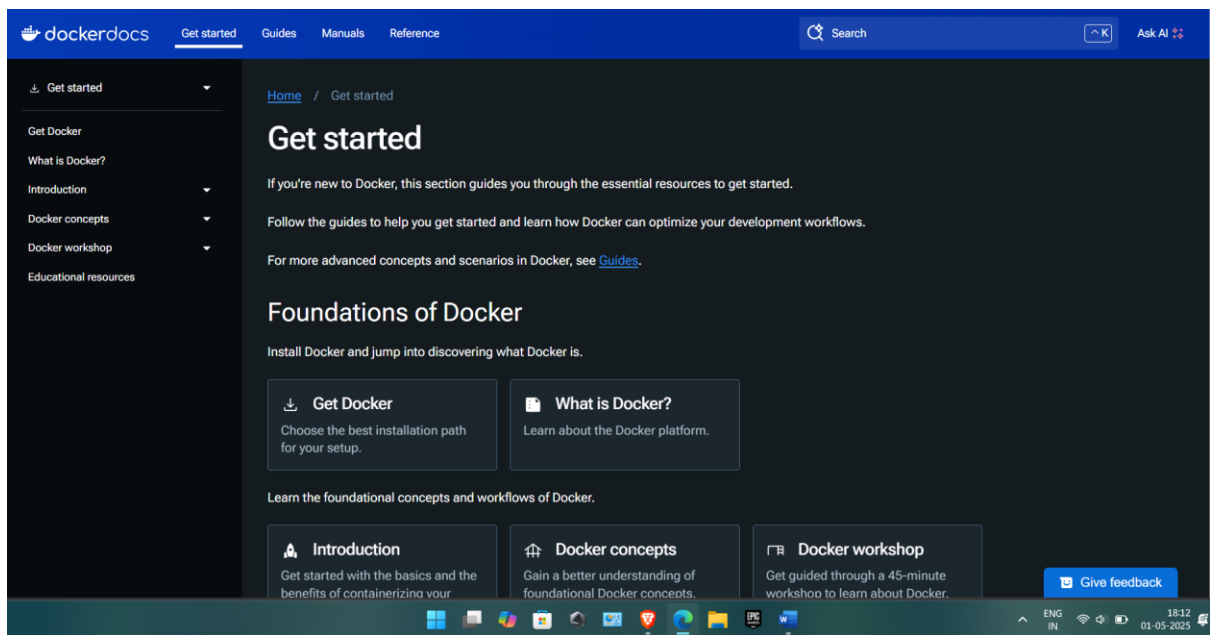
To try something more ambitious, you can run an Ubuntu container with:
\$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
<https://hub.docker.com/>

For more examples and ideas, visit:

i-07a117e4467ac87de (kube-instance)

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```
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```

```
500.30 MiB / 500.31 MiB 100.00% 65.80 M
* Creating docker container (CPUs=2, Memory=2200MB) ...
* Preparing Kubernetes v1.32.0 on Docker 27.4.1 ...
  - Generating certificates and keys ...
  - Booting up control plane ...
  - Configuring RBAC rules ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@ip-172-31-14-54:~$ kubectl get nodes
NAME               STATUS   ROLES    AGE   VERSION
minikube            Ready    control-plane 11s   v1.32.0
ubuntu@ip-172-31-14-54:~$ kubectl create namespace test-ns
namespace/test-ns created
ubuntu@ip-172-31-14-54:~$ kubectl get namespaces
NAME                STATUS   AGE
default              Active   35s
kube-node-lease      Active   35s
kube-public          Active   35s
kube-system          Active   35s
test-ns              Active   14s
ubuntu@ip-172-31-14-54:~$ kubectl config set-context --current --namespace=test-ns
Context "minikube" modified.
ubuntu@ip-172-31-14-54:~$ kubectl run nginx --image=nginx --port=80
pod/nginx created
ubuntu@ip-172-31-14-54:~$ kubectl get pods
NAME    READY   STATUS    RESTARTS   AGE
nginx   1/1     Running   0           11s
```

i-07a117e4467ac87de (kube-instance)

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```
ubuntu@ip-172-31-14-54:~$ kubectl run nginx --image=nginx --port=80
pod/nginx created
ubuntu@ip-172-31-14-54:~$ kubectl get pods
NAME    READY   STATUS    RESTARTS   AGE
nginx   1/1     Running   0           11s
ubuntu@ip-172-31-14-54:~$ kubectl get namespaces
NAME                STATUS   AGE
default              Active   102s
kube-node-lease      Active   102s
kube-public          Active   102s
kube-system          Active   102s
test-ns             Active    81s
ubuntu@ip-172-31-14-54:~$ kubectl run nginx --image=nginx --namespace=demo-ns
Error from server (NotFound): namespaces "demo-ns" not found
ubuntu@ip-172-31-14-54:~$ kubectl run nginx --image=nginx --namespace=test-ns
Error from server (AlreadyExists): pods "nginx" already exists
ubuntu@ip-172-31-14-54:~$ kubectl delete namespace test-ns
namespace "test-ns" deleted
^[A*[[A
ubuntu@ip-172-31-14-54:~$ kubectl run nginx --image=nginx --namespace=test-ns
Error from server (NotFound): namespaces "test-ns" not found
ubuntu@ip-172-31-14-54:~$ kubectl get namespaces
NAME                STATUS   AGE
default              Active   3m34s
kube-node-lease      Active   3m34s
kube-public          Active   3m34s
kube-system          Active   3m34s
ubuntu@ip-172-31-14-54:~$
```

OVERALL COMMANDS:

```
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```
default              Active   3m34s
kube-node-lease      Active   3m34s
kube-public          Active   3m34s
kube-system          Active   3m34s
ubuntu@ip-172-31-14-54:~$ history
1  docker run hello-world
2  curl -LO "https://dl.k8s.io/release/${curl -sL https://dl.k8s.io/release/stable.txt}/bin/linux/amd64/kubectl"
3  chmod +x kubectl
4  sudo mv kubectl /usr/local/bin/
5  kubectl version --client
6  curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
7  chmod +x minikube-linux-amd64
8  sudo mv minikube-linux-amd64 /usr/local/bin/minikube
9  minikube version
10 minikube start --driver=docker
11 kubectl get nodes
12 kubectl create namespace test-ns
13 kubectl get namespaces
14 kubectl config set-context --current --namespace=test-ns
15 kubectl run nginx --image=nginx --port=80
16 kubectl get pods
17 kubectl get namespaces
18 kubectl run nginx --image=nginx --namespace=demo-ns
19 kubectl run nginx --image=nginx --namespace=test-ns
20 kubectl delete namespace test-ns
21 kubectl run nginx --image=nginx --namespace=test-ns
22 kubectl get namespaces
23 history
ubuntu@ip-172-31-14-54:~$
```

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